**Math 2320 - FE 1.2 Review**

1. $\int \sin^2(3x)\cos(3x)dx$

2. $\int \sin^2(3x)\cos^3(3x)dx$

3. $\int \sin^2(3x)dx$

4. $\int \sin^2(4x)\cos^2(4x)dx$

5. $\int \tan^3(x)\sec^2(x)dx$

6. $\int \frac{1}{\sqrt{4-9x^2}}dx$

7. $\int \frac{1}{4+9x^2}dx$

8. $\int \frac{1}{(1-x^2)^{3/2}}dx$

9. $\int \frac{1}{x\sqrt{1+x^2}}dx$

10. $\int \frac{1}{x^2\sqrt{1-x^2}}dx$

11. What are the radii and intervals of convergence of the following power series.

   (a) $\sum_{n=1}^{\infty} \frac{1}{n} x^n$

   (b) $\sum_{n=1}^{\infty} \frac{1}{n^2} x^n$

   (c) $\sum_{n=1}^{\infty} \frac{1}{n^2} (x-5)^n$

   (d) $\sum_{n=0}^{\infty} \frac{1}{n!} x^n$

   (e) $\sum_{n=0}^{\infty} x^n$

   (f) $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$

12. Find the Taylor series from the definition

   (a) $f(x) = \sin(2x)$ at $x = 0$

   (b) $f(x) = e^x$ at $x = 0$

   (c) $f(x) = \frac{1}{1-x}$ at $x = 0$

   (d) $f(x) = \ln(x+1)$ at $x = 0$